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Introduction

Artificial intelligence systems are informed by the structure and function of biological neuron systems. However, while artificial neuron networks borrow conceptual ideas from neuroscience, the mechanisms used in these networks differ significantly from those found in biological neurons. The purpose of this lab was to simulate and compare biological neuron behavior and artificial neuron models in order to better understand how neural signals are processed, how activation functions change results, and how insights from neuroscience may change AI architecture.

Findings

Biological and artificial neurons both transform input signals into output responses, but they do so through fundamentally different means. Biological neurons rely upon membrane voltage dynamics, threshold-triggered action potentials, refractory periods, and time-dependent electrical behavior to process information. When sufficient input causes the membrane potential to exceed a threshold, the neuron generates an action potential, called a spike, which propagates the signal to other neurons. Artificial neurons, however, perform a computation of a weighted sum of inputs and apply an activation function. In our lab activity, we utilized the activation functions of sigmoid, rectified linear unit (ReLU), and hyperbolic tangent (tanh).

While both systems exhibit threshold-like behavior, biological neurons operate through dynamic, spike-based signaling over time, whereas artificial neurons represent simplified, continuous mathematical answers.

In the biological neuron simulation, stronger input currents caused the neuron to reach its firing threshold more often, producing a higher rate of “spikes” over time. When the input remained too weak, the membrane potential stayed below the firing trigger point, and no action potential occurred. This demonstrates the all-or-none principle of biological neurons, where action potentials occur only when the membrane potential crosses a specific threshold. The simulation also illustrates that biological neurons are not purely linear devices; instead, they integrate incoming signals over time, respond selectively to sufficient stimulation, and are constrained by refractory periods that temporarily prevent additional spikes. These dynamics shape firing patterns and affect how information is moved through neural circuits.

The artificial neuron simulation demonstrated that neuron behavior varies depending upon the activation function applied to the weighted input sum. The sigmoid function compresses outputs into a smooth range between 0 and 1, the tanh function produces outputs between -1 and 1, and the ReLU function passes through positive values while setting negative values to zero.

These activation functions introduce nonlinearity into artificial neural networks and act similarly to biological thresholds. Unlike biological thresholds, however, they remain simplified approximations of real neural behavior. And these mathematical functions do not represent spike timing, refractory periods, or membrane potential dynamics. These are key elements of biological information processing.

Both biological and artificial neuron networks demonstrate that the structure of a network significantly influences how information is processed. In the simple network simulations, different combinations of inputs produced distinct output patterns, and even small changes in network architecture affected the resulting behavior.

Implications for AI

Observations from the networks can be used for improving AI systems. Biological neurons operate with energy efficiency, adapt dynamically to changing inputs, and process information in event-driven ways that depend upon spike timing and temporal integration. These properties have inspired emerging AI paradigms such as spiking neural networks, neuromorphic computing, and architectures designed to incorporate temporal dynamics and sparse activation patterns. While current ANNs have achieved significant success, they remain simplistic models of biological systems. Using biological principles related to timing, memory integration, and adaptive signaling could lead to more efficient and biologically inspired AI architectures.

Limitations & Potential Improvements

Despite their usefulness, ANNs omit many essential features of biological neural systems. Artificial neurons typically ignore membrane potential dynamics, spike timing behavior, neuromodulation, refractory mechanisms, and biochemical plasticity that influences learning and adaptation in biological neurons. As a result, these models function primarily as engineering abstractions rather than biologically faithful representations. Future improvements could include incorporating temporal coding mechanisms, adaptive thresholds, event-driven computation, and closer integration between memory and learning processes.